

Forensic Examination in the matter of

The cold case investigation of Malcom Monarch, Monarch laptop and Clarion Hotel kiosk

Overview

Detective Angela Lansbury referred this cold case on the activity of supervillain Malcom Monarch. The Monarch dropped his broken laptop at Conjectural Technologies for repair, their technician, Pete White, spotted suspicious activity and with a warrant acquired the laptop for law enforcement. A second image, a Clarion Hotel internet kiosk believed used by the Monarch while on the run, was provided on consent. Agent Sampson asked me to pull communications, web searches, and media related to the Monarch's activity, a North Carolina Superior Court warrant covers the request.

Working primarily in Autopsy, I examined the Monarch laptop image and its registry hives. The exam recovered the Monarch's email with a self identified Guild operative, a web search trail showing a plan to attack Dr. Venture with a weaponized drone and explosives, searches and tools aimed at hiding evidence and defeating forensics and downloaded photographs of the subject that he also opened locally in a browser. Each finding below answers one of Agent Sampson's questions, with the Autopsy artifact it came from.

Two things to keep in mind for every timestamp in this report. The subject's system time zone is Pacific some Autopsy columns render in Eastern because of the case time zone setting, so each time is labeled with its zone. And the browser history records the subject's profile path as IEUser, the same filenames resolve to the files present in Malcom's user folder.

Exam Preparation

Two logical E01 images were provided, the Monarch laptop and the Clarion kiosk each 5000 MB. Each stores its own MD5 and SHA1, and I verified both against their acquisition logs; both verify.

Image	Case / acquired	MD5	Result
MonarchIMG2021-1.E01	Lab 5 / 2021-03-30	dd2ddf067505a2a35264528dc43dc448	Verified
ClarionBusinessCenter.E01	TheMonarch / 2024-02-21	81c5fe2491a0bfdd66aaf3c64d4886a1	Verified

Accounting for all the space: total size = bytes per sector × sector count = $512 \times 10,240,000 = 5,242,880,000$ bytes, 5000 MB, matching the source size in each acquisition log, so each image accounts for the whole device including unallocated space.

Source file metadata, Autopsy and Monarch image

Autopsy's Source File Metadata for the Monarch image confirms the key acquisition facts in one place:

Field	Value
Name	MonarchIMG2021-1.E01
Type / Size	E01 · 5,242,880,000 bytes (5000 MB)
Sector Size	512 bytes
MD5	dd2ddf067505a2a35264528dc43dc448
SHA1	9d1553b3a4d5e13002a659d0c9f32545fbc3b136
Time Zone	America/Los_Angeles (Pacific)
Acquired	2021-03-31 02:12:15 (per acquisition log) · ADI 4.5.0.3

Tools

Primary analysis in Autopsy 4.23.1 on Windows, ingest: Email Parser, Recent Activity/web, Extension Mismatch, Picture Analyzer/EXIF, PhotoRec Carver, Keyword Search, Hash Lookup. Registry hives SAM, SECURITY, SOFTWARE, SYSTEM from RegistryM/RegistryC were read for OS version and time zone. Findings were cross checked with The Sleuth Kit, libewf (ewfverify), and ESE/mbox parsing of the mail and WebCache; both paths agreed.

Time zone basis

The SYSTEM hive's TimeZoneInformation on both machines is Pacific Standard Time (Bias 480 = UTC-8; ActiveTimeBias 420 = UTC-7 during daylight. Email headers and Windows FILETIMES are stored in UTC; EXIF and some artifacts are local. Where Autopsy columns display Eastern it is a case-setting artifact, not the system's zone, each time below is labeled.

Analysis

1) Was there communication from someone claiming to be the Guild's operative, or can you identify the operative's name/email?

Yes. Recovered email shows a person writing on behalf of the Guild of Calamitous Intent. He identifies himself in his signature as Hamilton J. Fantomos, alias The Phantom Limb, emailing from generalconsolidatedindustries@aol.com, fronting as General Consolidated Industries. On 2015-04-10 he emailed the Monarch (mightymonarch@outlook.com), subject Your Application, approving his application for reinstatement as a super villain and granting arching rights against Dr. Venture, signing Hamilton J Fantomos, The Phantom Limb, Accountant, Insurance Agent, Supervillain.

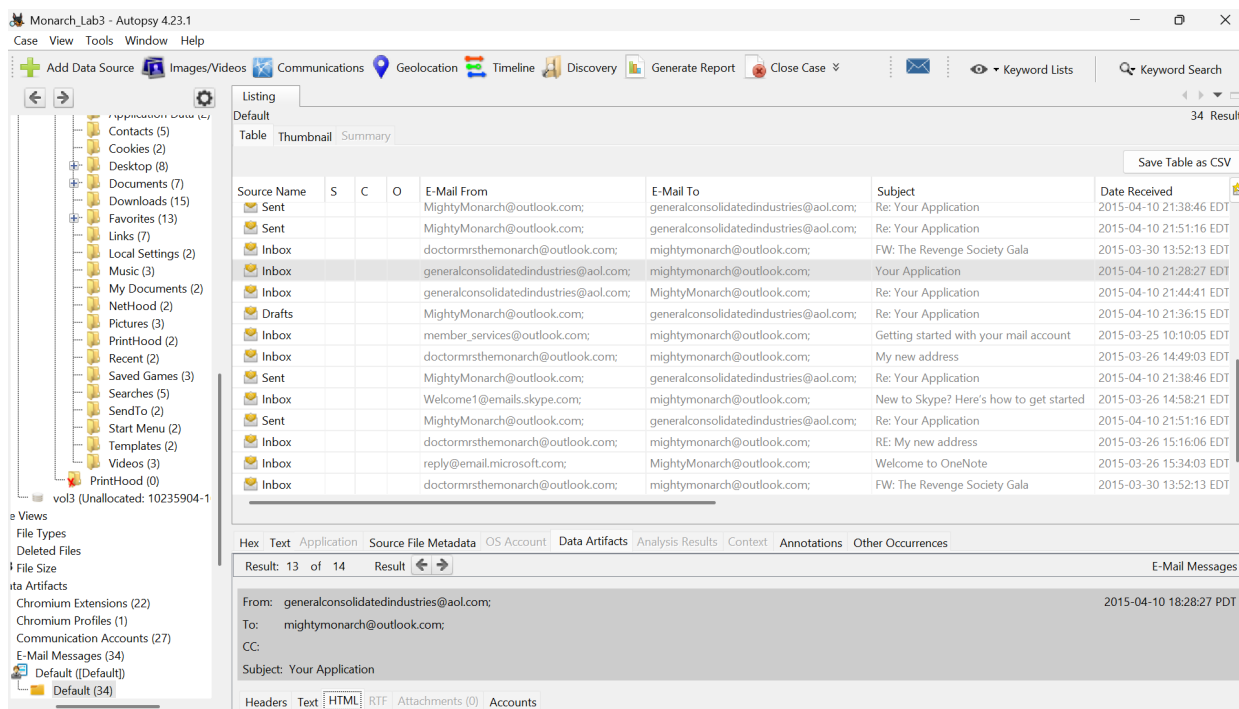


Figure 1: Autopsy — E-Mail Messages: operative email from generalconsolidatedindustries@aol.com

2) What is Malcom's email address? Did he send email to someone other than the guild operative? Who?

Malcom's email address is mightymonarch@outlook.com he also used a second account, doctormrsthemonarch@outlook.com). Yes, besides the operative, he emailed his wife, Sheila Monarch (doctormrsthemonarch@outlook.com), in threads such as My new address, Im Famous, Recruitment, and FW:

The Revenge Society Gala. The remaining senders, Outlook, Skype, Microsoft are automated service mail, not correspondents.

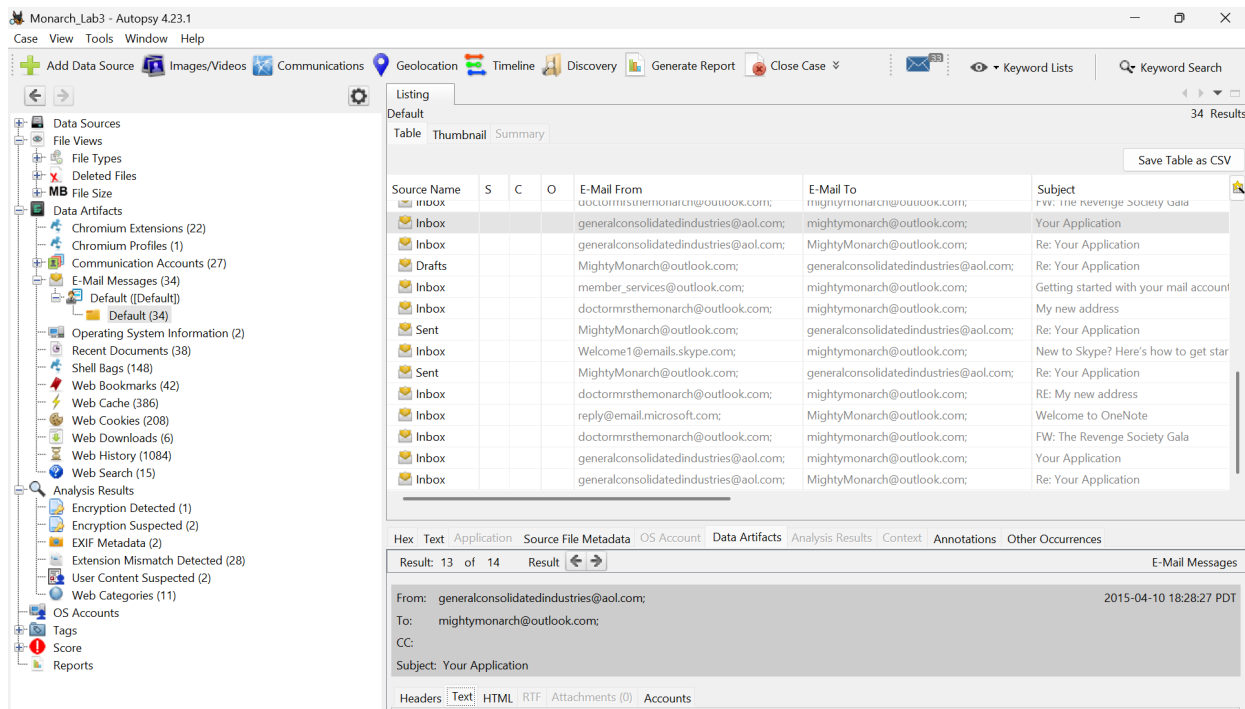


Figure 2: Autopsy — E-Mail Messages list showing Malcom's addresses and his correspondents.

3) What version OS was running on the computer in use?

Windows 7 Enterprise, build 7601 version 6.1, installed 2013-10-23. This is read from the SOFTWARE hive key Microsoft\Windows NT\CurrentVersion and confirmed in Autopsy's Operating System Information. Both the Monarch and Clarion images report the same OS build.

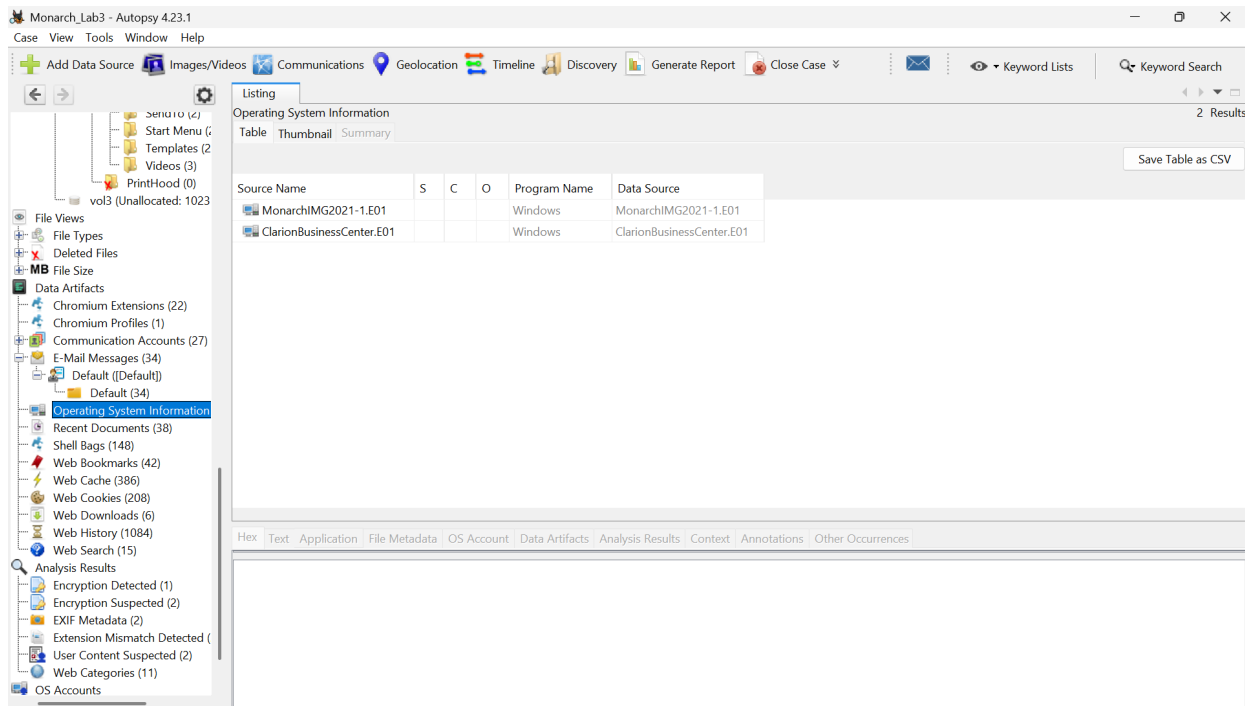
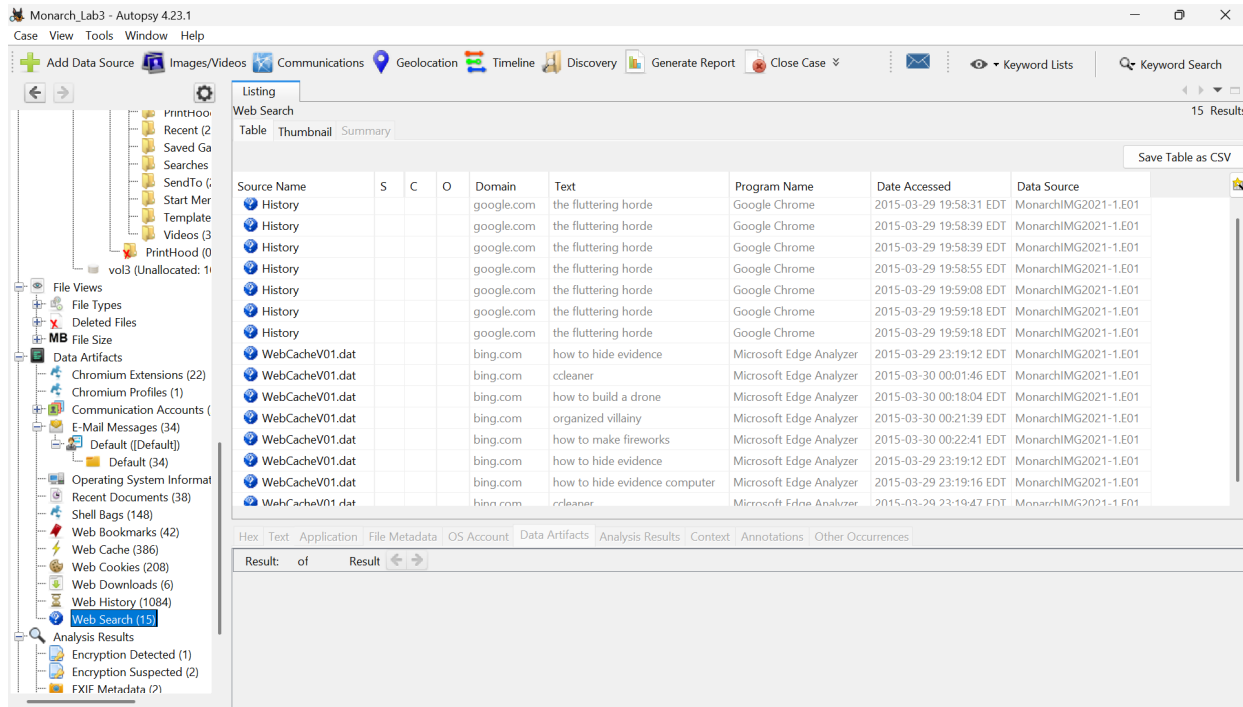


Figure 3: Autopsy — Operating System Information

4) Did Malcon ever search for any terms that might mean he was going to cause any explosions?

Yes. Recovered Bing searches include how to make fireworks and how to build a drone, and his browser history shows Wikipedia visits to Pipe bomb and Bomb making instructions on the internet. These line up with his own email to the operative, I'm going to poison Venture, then blow him up with a robot drone and with drone shopping Amazon, Gilt, HSN, myfirstdrone.com and the search kill a man with a drone. Together this indicates a plan to use a weaponized drone and/or explosives against Dr. Thaddeus Rusty Venture.



Source Name	S	C	O	Domain	Text	Program Name	Date Accessed	Data Source
History				google.com	the fluttering horde	Google Chrome	2015-03-29 19:58:31 EDT	MonarchIMG2021-1.E01
History				google.com	the fluttering horde	Google Chrome	2015-03-29 19:58:39 EDT	MonarchIMG2021-1.E01
History				google.com	the fluttering horde	Google Chrome	2015-03-29 19:58:39 EDT	MonarchIMG2021-1.E01
History				google.com	the fluttering horde	Google Chrome	2015-03-29 19:58:55 EDT	MonarchIMG2021-1.E01
History				google.com	the fluttering horde	Google Chrome	2015-03-29 19:59:08 EDT	MonarchIMG2021-1.E01
History				google.com	the fluttering horde	Google Chrome	2015-03-29 19:59:18 EDT	MonarchIMG2021-1.E01
History				google.com	the fluttering horde	Google Chrome	2015-03-29 19:59:18 EDT	MonarchIMG2021-1.E01
WebCacheV01.dat				bing.com	how to hide evidence	Microsoft Edge Analyzer	2015-03-29 23:19:12 EDT	MonarchIMG2021-1.E01
WebCacheV01.dat				bing.com	ccleaner	Microsoft Edge Analyzer	2015-03-30 00:01:46 EDT	MonarchIMG2021-1.E01
WebCacheV01.dat				bing.com	how to build a drone	Microsoft Edge Analyzer	2015-03-30 00:18:04 EDT	MonarchIMG2021-1.E01
WebCacheV01.dat				bing.com	organized villainy	Microsoft Edge Analyzer	2015-03-30 00:21:39 EDT	MonarchIMG2021-1.E01
WebCacheV01.dat				bing.com	how to make fireworks	Microsoft Edge Analyzer	2015-03-30 00:22:41 EDT	MonarchIMG2021-1.E01
WebCacheV01.dat				bing.com	how to hide evidence	Microsoft Edge Analyzer	2015-03-29 23:19:12 EDT	MonarchIMG2021-1.E01
WebCacheV01.dat				bing.com	how to hide evidence computer	Microsoft Edge Analyzer	2015-03-29 23:19:16 EDT	MonarchIMG2021-1.E01
WebCacheV01.dat				bing.com	ccleaner	Microsoft Edge Analyzer	2015-03-29 23:19:47 EDT	MonarchIMG2021-1.E01

Figure 4: Autopsy — Web Search list including “how to make fireworks” and “how to build a drone.”

5) How about how to cover his tracks?

Yes. He searched how to hide evidence and how to hide evidence computer, read the Wikipedia article Anti computer forensics, and searched for and downloaded CCleaner, ccsetup504.exe is present in his Downloads folder, and a tool used to wipe files and activity. This shows deliberate intent to remove or conceal evidence on the machine.

Source Name	S	C	O	Domain	Text	Program Name	Date Accessed	Data Source
History				google.com	the fluttering horde	Google Chrome	2015-03-29 19:58:31 EDT	MonarchIMG2021-1.E01
History				google.com	the fluttering horde	Google Chrome	2015-03-29 19:58:39 EDT	MonarchIMG2021-1.E01
History				google.com	the fluttering horde	Google Chrome	2015-03-29 19:58:39 EDT	MonarchIMG2021-1.E01
History				google.com	the fluttering horde	Google Chrome	2015-03-29 19:58:55 EDT	MonarchIMG2021-1.E01
History				google.com	the fluttering horde	Google Chrome	2015-03-29 19:59:08 EDT	MonarchIMG2021-1.E01
History				google.com	the fluttering horde	Google Chrome	2015-03-29 19:59:18 EDT	MonarchIMG2021-1.E01
History				google.com	the fluttering horde	Google Chrome	2015-03-29 19:59:18 EDT	MonarchIMG2021-1.E01
WebCacheV01.dat				bing.com	how to hide evidence	Microsoft Edge Analyzer	2015-03-29 23:19:12 EDT	MonarchIMG2021-1.E01
WebCacheV01.dat				bing.com	ccleaner	Microsoft Edge Analyzer	2015-03-30 00:01:46 EDT	MonarchIMG2021-1.E01
WebCacheV01.dat				bing.com	how to build a drone	Microsoft Edge Analyzer	2015-03-30 00:18:04 EDT	MonarchIMG2021-1.E01
WebCacheV01.dat				bing.com	organized villainy	Microsoft Edge Analyzer	2015-03-30 00:21:39 EDT	MonarchIMG2021-1.E01
WebCacheV01.dat				bing.com	how to make fireworks	Microsoft Edge Analyzer	2015-03-30 00:22:41 EDT	MonarchIMG2021-1.E01
WebCacheV01.dat				bing.com	how to hide evidence	Microsoft Edge Analyzer	2015-03-29 23:19:12 EDT	MonarchIMG2021-1.E01
WebCacheV01.dat				bing.com	how to hide evidence computer	Microsoft Edge Analyzer	2015-03-29 23:19:16 EDT	MonarchIMG2021-1.E01
WebCacheV01.dat				bing.com	ccleaner	Microsoft Edge Analyzer	2015-03-29 23:19:47 EDT	MonarchIMG2021-1.E01

Figure 5: Autopsy — Web Search list: “how to hide evidence” / ”how to hide evidence computer” and “ccleaner.”

6) Explain in detail what he was doing, with what software and when.

All times Pacific unless noted, Autopsy displayed several columns in Eastern, so subtract 3 hours for local, or read as labeled. In late March 2015 Malcom Monarch set up a new Windows 7 Enterprise workstation and, using Internet Explorer and Windows Live Mail, carried out the following:

- Mar 26, 2015, emailed his wife Sheila about his “new address,” being on the road with henchmen, and setting up the new workstation.
- Mar 29–30, 2015, using Internet Explorer, searched Bing for “how to build a drone,” “how to make fireworks,” and “kill a man with a drone”; shopped quadcopter drones (Amazon/Gilt/HSN/myfirstdrone.com), read Wikipedia on pipe bombs and bomb making, building a plan to attack Dr. Venture.
- Mar 29–30, 2015, downloaded Monarch images to his Downloads folder and opened them locally in the browser (file:///).
- Mar 29–30, 2015, searched “how to hide evidence” read “Anti-computer forensics,” visited Magnet Forensics and AccessData, and downloaded CCleaner (ccsetup504.exe) to cover his tracks.
- Mar 29 – Apr 10, 2015, corresponded with Guild operative Hamilton Fantomos (“The Phantom Limb”) about his villain “application,” replied threatening to “poison Venture, then blow him up with a robot drone,” and exchanged mail with his wife Sheila.

Software used: Internet Explorer, Windows Live Mail, CCleaner

7) Are there any photos that might have been downloaded? How do you know the photo was downloaded and not just moved to the folder where it was found? Is there a photo at this location that might tell us how he appears?

Yes. In C:\Users\Malcom\Downloads there are two photographs of the subject:

the_monarch_wants_you__again_by_petex-d2yvilyz.jpg (a “THE MONARCH WANTS YOU!” recruitment poster) and venture-bros-monarch.jpg (a portrait of the Monarch in costume). Both show how the subject appears — a man in a yellow-and-black monarch-butterfly supervillain costume with antennae.

I can tell these were downloaded, not moved, from several converging signs, they sit in the browser’s default Downloads folder; the filename the_monarch_wants_you__again_by_petex-d2yvilyz.jpg carries a deviantART

work ID (“petex-d2yviltz”), i.e. the originating site’s name rather than a locally authored name; their Created and Accessed times are within a few seconds of each other written once, as a download is, not spread out as edited local files would be, and matching .lnk shortcuts under AppData\...\Recent record that they were opened. The browser history also records these files being opened from the Downloads path.



Figure 6: the monarch wants_you_by_petex-d2yviltz.jpg, recovered from Malcom’s Downloads (how the subject appears).

Monarch_Lab3 - Autopsy 4.23.1

Case View Tools Window Help

Add Data Source Images/Videos Communications Geolocation Timeline Discovery Generate Report Close Case

Keyword Lists Keyword Search

Listing

/img_MonarchIMG2021-1.E01/vol_vol2/Users/Malcom/Downloads 15 Results

Name	S	C	O	Modified Time	Change Time	Access Time	Created Time	Size
[parent folder]				2021-03-30 21:55:42 EDT	2021-03-30 21:55:42 EDT	2021-03-30 21:55:42 EDT	2021-03-30 21:54:20 EDT	56
\$130			0	2015-03-29 19:59:31 EDT	2021-03-24 21:30:28 EDT	2015-03-29 19:59:31 EDT	2013-10-23 12:22:47 EDT	4096
Build Your Own Drone _Flight Today _Air & Space			0	2015-04-11 11:09:18 EDT	2021-03-24 21:31:10 EDT	2015-04-11 11:09:18 EDT	2015-04-11 11:09:17 EDT	716942
Build Your Own Drone _Flight Today _Air & Space			0	2021-03-24 21:31:10 EDT	2021-03-24 21:31:10 EDT	2021-03-24 21:31:10 EDT	2021-03-24 21:31:10 EDT	3954
ccsetup504.exe			0	2015-03-29 19:39:29 EDT	2021-03-24 21:30:28 EDT	2015-03-29 19:39:29 EDT	2015-03-29 19:39:58 EDT	5344528
desktop.ini			1	2021-03-30 21:55:38 EDT	2021-03-30 21:55:38 EDT	2021-03-30 21:55:38 EDT	2013-10-23 12:23:07 EDT	282
images (2).jpg			0	2015-04-11 11:08:50 EDT	2021-03-24 21:31:10 EDT	2015-04-11 11:08:49 EDT	2015-04-11 11:08:50 EDT	4472
images (2).jpg.FileSlack			0	2021-03-24 21:31:10 EDT	2021-03-24 21:31:10 EDT	2021-03-24 21:31:10 EDT	2021-03-24 21:31:10 EDT	3720
images.jpg			1	2015-03-29 19:59:05 EDT	2021-03-24 21:30:28 EDT	2015-03-29 19:59:02 EDT	2015-03-29 19:59:05 EDT	14174
images.jpg.FileSlack			0	2021-03-24 21:30:28 EDT	2021-03-24 21:30:28 EDT	2021-03-24 21:30:28 EDT	2021-03-24 21:30:28 EDT	2210
the_monarch_wants_you_again_by_petex-d2yviltz.jpg			1	2015-03-29 19:58:54 EDT	2021-03-24 21:30:28 EDT	2015-03-29 19:58:51 EDT	2015-03-29 19:58:54 EDT	259797
the_monarch_wants_you_again_by_petex-d2yviltz.jpg			0	2021-03-24 21:30:28 EDT	2021-03-24 21:30:28 EDT	2021-03-24 21:30:28 EDT	2021-03-24 21:30:28 EDT	2347
venture-bros-monarch.jpg			1	2015-03-29 19:59:31 EDT	2021-03-24 21:30:28 EDT	2015-03-29 19:59:29 EDT	2015-03-29 19:59:31 EDT	242490
venture-bros-monarch.jpg.FileSlack			0	2021-03-24 21:30:28 EDT	2021-03-24 21:30:28 EDT	2021-03-24 21:30:28 EDT	2021-03-24 21:30:28 EDT	3270

Save Table as CSV

Hex Text Application File Metadata OS Account Data Artifacts Analysis Results Context Annotations Other Occurrences

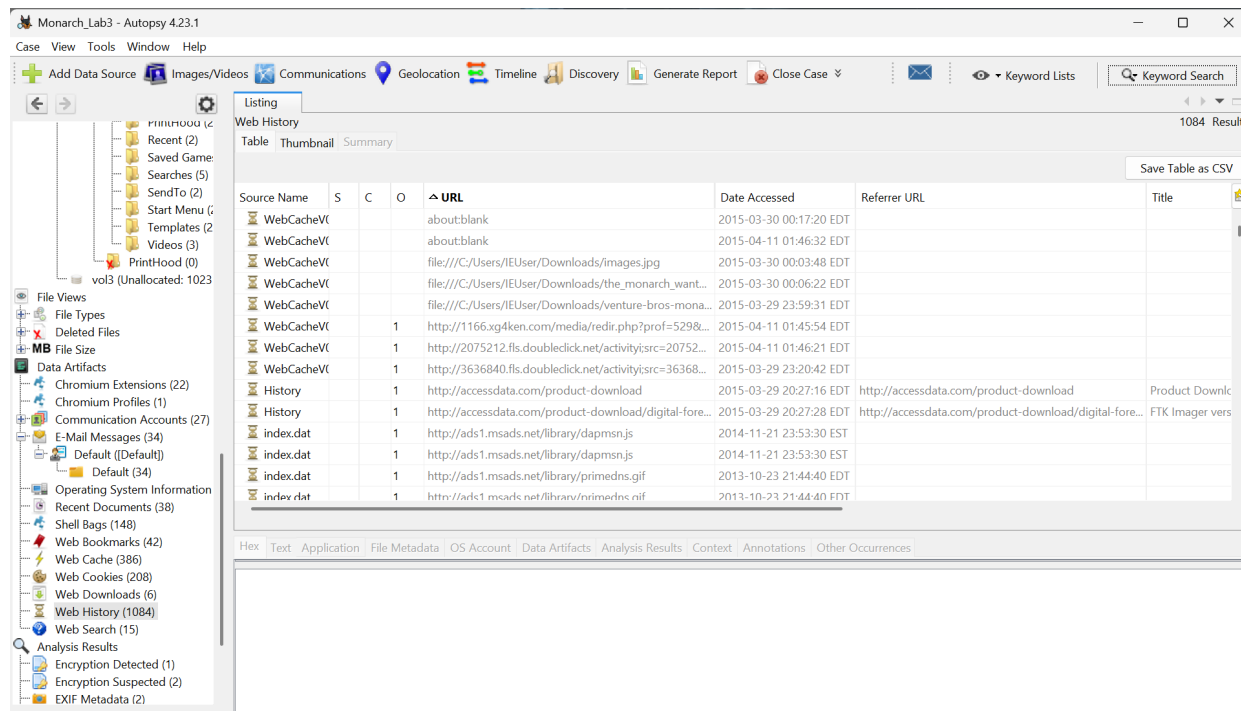
0% 17% Reset

Tags Menu

Figure 7: Autopsy — Users\Malcom\Downloads listing (poster, portrait, ccsetup504.exe) with image preview.

8) Did malcom do any research into forensics software? What specifically?

Yes. His history shows research into three named tools: Magnet Forensics (magnetforensics.com, including a page titled “how to uncover the covered tracks / hidden evidence using computer forensics tools”), AccessData FTK Imager (accessdata.com/product-download — history title “FTK Imager version...”), and CCleaner (searched and downloaded ccsetup504.exe). In short, he researched the very forensic and anti-forensic tools that would be used to examine, or to wipe, his machine.



The screenshot shows the Autopsy 4.23.1 interface with the 'Web History' table selected. The table lists various web history entries, including several visits to 'accessdata.com/product-download' and 'magnetforensics.com'. The 'Source Name' column shows 'WebCacheV' for most entries, while 'History' and 'index.dat' are also present. The 'URL' column contains the specific web addresses accessed. The 'Date Accessed' column shows the timestamps. The 'Referrer URL' column shows the source of the traffic. The 'Title' column shows the page titles. The table is sorted by 'Date Accessed' in descending order.

Source Name	S	C	O	URL	Date Accessed	Referrer URL	Title
WebCacheV				about:blank	2015-03-30 00:17:20 EDT		
WebCacheV				about:blank	2015-04-11 01:46:32 EDT		
WebCacheV				file:///C:/Users/IEUser/Downloads/images.jpg	2015-03-30 00:03:48 EDT		
WebCacheV				file:///C:/Users/IEUser/Downloads/the_monarch_want...	2015-03-30 00:06:22 EDT		
WebCacheV				file:///C:/Users/IEUser/Downloads/venture-bros-mona...	2015-03-29 23:59:31 EDT		
WebCacheV			1	http://1166.xg4ken.com/media/edir.php?prof=5298...	2015-04-11 01:45:54 EDT		
WebCacheV			1	http://2075212.fls.doubleclick.net/activitysrc=20752...	2015-04-11 01:46:21 EDT		
WebCacheV			1	http://3636840.fls.doubleclick.net/activitysrc=36368...	2015-03-29 23:20:42 EDT		
History			1	http://accessdata.com/product-download	2015-03-29 20:27:16 EDT	http://accessdata.com/product-download	Product Downl...
History			1	http://accessdata.com/product-download/digital-fore...	2015-03-29 20:27:28 EDT	http://accessdata.com/product-download/digital-fore...	FTK Imager vers...
index.dat			1	http://ads1.msads.net/library/dapmsn.js	2014-11-21 23:53:30 EST		
index.dat			1	http://ads1.msads.net/library/dapmsn.js	2014-11-21 23:53:30 EST		
index.dat			1	http://ads1.msads.net/library/primedns.gif	2013-10-23 21:44:40 EDT		
index.dat			1	http://ads1.msads.net/library/primedns.gif	2013-10-23 21:44:40 EDT		

Figure 8: Autopsy — Web History: accessdata.com (FTK Imager) research rows.

9) Is there any evidence he may have opened photos locally with an internet browser? Are those photos still available on the system?

Yes. The browser history contains three file:/// entries — local files opened in the browser rather than web pages: file:///C:/Users/IEUser/Downloads/images.jpg, .../the_monarch_wants_you_again_by_petex-d2yviltz.jpg, and .../venture-bros-monarch.jpg (accessed 2015-03-29 to 03-30). A file:/// URL only appears when a local file is opened in the browser. And yes — those photos are still on the system; they are the same files present in the Downloads folder (Q7). The “IEUser” path is the built-in test-build profile; the filenames resolve to the files in Malcom’s Downloads.

Source Name	S	C	O	URL	Date Accessed	Referrer URL	Title
WebCacheV				about:blank	2015-03-30 00:17:20 EDT		
WebCacheV				about:blank	2015-04-11 01:46:32 EDT		
WebCacheV				file:///C:/Users/IEUser/Downloads/images.jpg	2015-03-30 00:03:48 EDT		
WebCacheV				file:///C:/Users/IEUser/Downloads/the_monarch_want...	2015-03-30 00:06:22 EDT		
WebCacheV				file:///C:/Users/IEUser/Downloads/venture-bros-mona...	2015-03-29 23:59:31 EDT		
WebCacheV			1	http://1166.xg4ken.com/media/edir.php?prof=5298...	2015-04-11 01:45:54 EDT		
WebCacheV			1	http://2075212.fls.doubleclick.net/activity;src=20752...	2015-04-11 01:46:21 EDT		
WebCacheV			1	http://3636840.fls.doubleclick.net/activity;src=36368...	2015-03-29 23:20:42 EDT		
History			1	http://accessdata.com/product-download	2015-03-29 20:27:16 EDT	http://accessdata.com/product-download	Product Downl...
History			1	http://accessdata.com/product-download/digital-fore...	2015-03-29 20:27:28 EDT	http://accessdata.com/product-download/digital-fore...	FTK Imager vers...
index.dat			1	http://ads1.msads.net/library/dapmsn.js	2014-11-21 23:53:30 EST		
index.dat			1	http://ads1.msads.net/library/dapmsn.js	2014-11-21 23:53:30 EST		
index.dat			1	http://ads1.msads.net/library/primedns.gif	2013-10-23 21:44:40 EDT		
index.rlat			1	http://ads1.msads.net/library/primedns.gif	2013-10-23 21:44:40 EDT		

Figure 9: Autopsy — Web History (sorted by URL): three file:/// local-photo opens, still present in Downloads.

Conclusion

The Monarch laptop image shows Malcom Monarch, in late March–April 2015, planning an attack on Dr. Thaddeus “Rusty” Venture using a weaponized drone and explosives, while deliberately researching how to hide evidence and defeat forensic tools. Recovered email establishes his contact with the Guild operative Hamilton Fantomos (“The Phantom Limb,” generalconsolidatedindustries@aol.com) approving his reinstatement, and his own written threat to “poison Venture, then blow him up with a robot drone.” His searches (fireworks, bombs, “kill a man with a drone,” “how to hide evidence”), his downloads (drone research, CCleaner, Monarch photos), and his research into Magnet Forensics and FTK Imager all corroborate intent and consciousness of guilt. Both evidence images verified against their stored hashes, and Autopsy and the command-line toolset returned the same findings, so I can testify to this examination.

Brian Tramuel
Signed B Tramuel

Appendix

Glossary

E01 image: A forensic copy of a drive that stores every bit of the original plus its own MD5 and SHA1, so the copy can be verified against the original.

Registry hive (SAM/SOFTWARE/SYSTEM): Windows configuration databases; SOFTWARE holds the OS version and install date, SYSTEM holds the time zone and device history.

WebCacheV01.dat: The Internet Explorer / WebCache ESE database that stores browsing history, searches, and downloads on Windows 7+.

file:/// URL: A browser address that points at a local file on the machine rather than a website; its presence in history means a local file was opened in the browser.

deviantART work ID: The code (e.g. “petex-d2yvzlz”) that deviantART appends to a downloaded image’s filename — a sign the file came from that site.

CCleaner: A cleanup tool often used to wipe files, histories, and traces — relevant here as an anti-forensic step.

Key artifacts referenced

- Email: generalconsolidatedindustries@aol.com (Hamilton Fantomos / Phantom Limb); mightymonarch@outlook.com and doctormrsthemonarch@outlook.com (Malcom / Sheila).
- Searches: how to build a drone, how to make fireworks, kill a man with a drone, how to hide evidence (computer), ccleaner.
- Pages: Wikipedia Pipe_bomb, Bomb-making_instructions_on_the_internet, Anti-computer_forensics; magnetforensics.com; accessdata.com (FTK Imager); piriform.com/ccleaner.
- Files: Users\Malcom\Downloads\{the_monarch_wants_you___again_by_petex-d2yvzlz.jpg, venture-bros-monarch.jpg, images.jpg, ccsetup504.exe}.